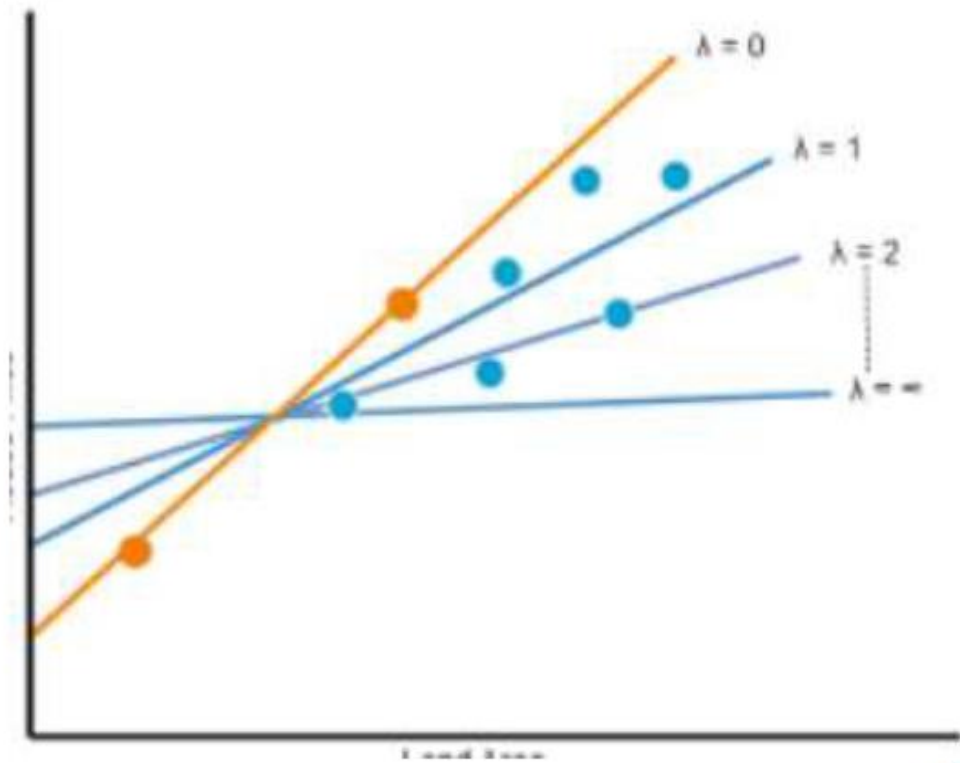
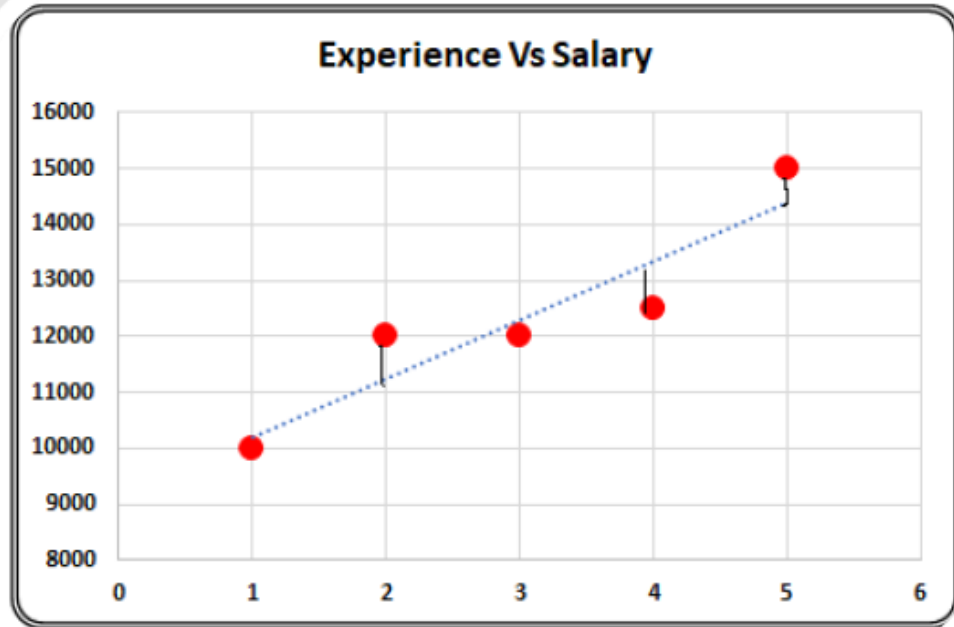


Lasso, Ridge & Elastic Net Regression



Linear Regression

Linear regression is an approach to model the relationship between a dependent variable and one or more independent variables.



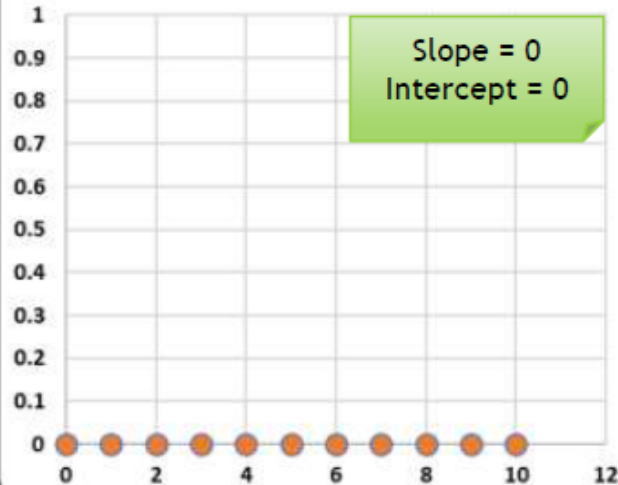
- Linear Regression model try to create a linear relationship between dependent (salary) and independent (experience) variables.
- It try to create the equation of a straight line (salary = $m * \text{experience} + b$) with minimum error (residual) between actual value and predicted value.

Linear Regression tries to create best fit line with minimum sum of residuals ($\sum (Y - Y_{\text{predict}})^2$) which is also known as cost function.

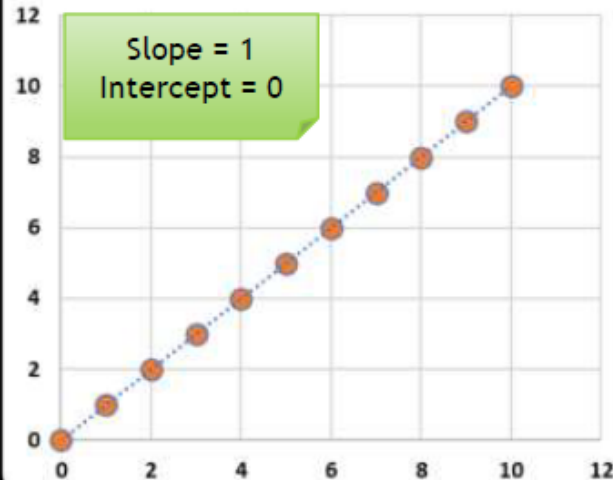
- Equation for simple linear regression : $y = m*x + b$
- Equation for multiple linear regression : $y = m_1*x_1 + m_2*x_2 + \dots + m_n*x_n + b$

Impact of slope (m)

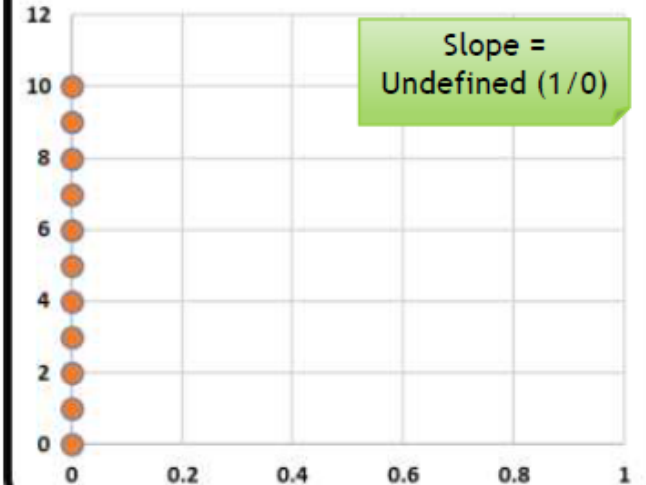
$$y = 0 \quad (y = 0 \cdot x + 0)$$



$$y = x \quad (y = 1 \cdot x + 0)$$



$$x = 0$$



- In case of the lines having higher slope, any minor variation in x can cause to a major variation in y .
- In the linear regression model creating with such data can have overfitting problem.
- One solution is to penalize the slopes and make the model a generalized one.
- LASSO and RIDGE regression are models will help for the same.

Underfitting occurs when a model can't accurately capture the dependencies among data, usually as a consequence of its own simplicity. It often yields a with known data and bad generalization applied with new data.

Overfitting happens when a model learns both dependencies among data and random words, a model learns the existing data too well. models usually yield high R^2 . However, they often generalize well and have significantly lower R^2 with new data.

Regularized Linear Model

A good way to reduce overfitting is to regularize the model (i.e., to constrain it):

For a linear model, regularization is typically achieved by constraining the weights of the model. Why?

The size of coefficients increase exponentially with increase in model complexity

Techniques like Ridge Regression, Lasso Regression, and Elastic Net, implement three different ways to constrain the weights.

Observations from Ridge and Lasso

Ridge and Lasso are linear regression techniques that add regularization to the cost function. Both aim to improve model performance and prevent overfitting.

Regularization helps by adding a penalty term based on the size of model coefficients.

- **Ridge Regression (L2 Regularization):** Slight reduction in slope magnitude due to the squared penalty term.
- **Lasso Regression (L1 Regularization):** Stronger penalty forces sparsity, with some coefficients set to zero.
- Both methods help improve model generalization.

RIDGE Regression

Ridge Regression (also called Tikhonov regularization) is a L2 regularized version of Linear Regression: a regularization term equal to $\alpha * \text{slope}^2$ is added to the cost function.

The minimization objectives is $\text{MSE} + \alpha * m^2$

This forces the learning algorithm to not only fit the data but also keep the model weights as small as possible.

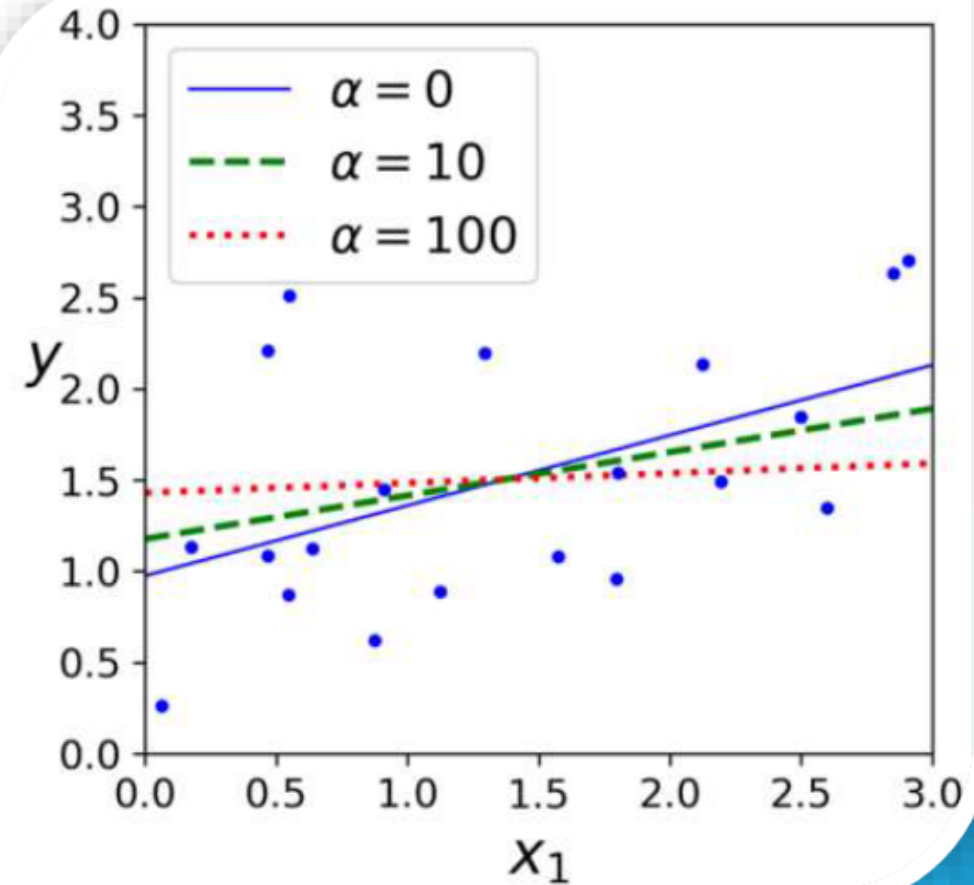
Note that the regularization term should only be added to the cost function during training.

The α (alpha) balance the amount of emphasis to the regularization. *Default value of alpha is 1. It can be zero to any positive number.*

$\alpha = 0 \rightarrow$ simple linear regression

$\alpha = \infty \rightarrow$ all coefficients become zero

increasing α leads to flatter (i.e., less extreme, more reasonable) predictions.



Ridge Regression

Key Idea: Ridge Regression minimizes residuals and adds a penalty on the sum of squared coefficients (θ_j^2).

Cost Function:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2 + \lambda \sum_{j=1}^n \theta_j^2$$

Where:

- λ controls the penalty strength.
- The penalty term shrinks coefficients, but never sets them to zero.

Ridge Regression - Effect of λ

- $\lambda = 0$: Equivalent to simple linear regression.
- $\lambda \rightarrow \infty$: Coefficients shrink to zero, leading to underfitting.

Example Data:

x	y
1	2
2	2.5
3	3.5

Ridge mitigates multicollinearity and reduces coefficient sizes.

Ridge Regression Example

Cost Function:

$$J(w_1, b) = \frac{1}{m} \sum_{i=1}^m \left(h(x^{(i)}) - y^{(i)} \right)^2 + \lambda w_1^2$$

Resulting Parameters:

$$w_1 = 0.5, \quad b = 1.67$$

Ridge Equation:

$$y = 0.5x + 1.67$$

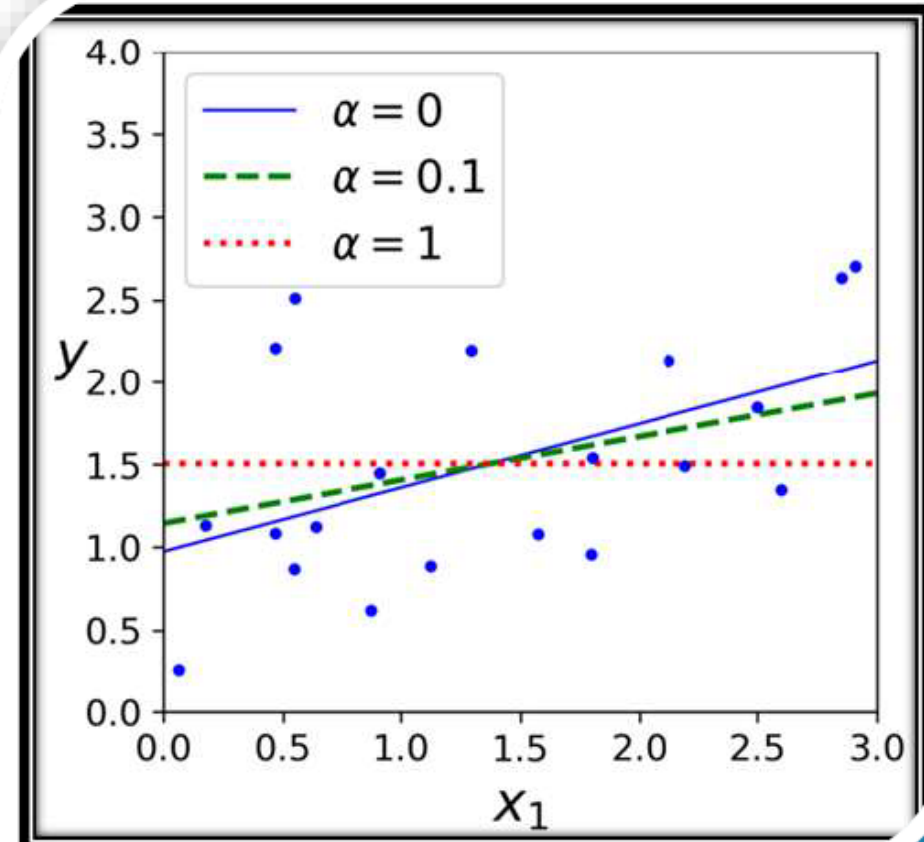
LASSO Regression

Least Absolute Shrinkage and Selection Operator Regression (simply called *Lasso Regression*) is another regularized version of Linear Regression: that adds a L1 regularization term to the cost function,

The minimization objectives is $\text{MSE} + \alpha * |m|$

Default value of alpha is 1. It can be zero to any positive number.

Lasso Regression tends to completely eliminate the weights of the least important features (i.e., set them to zero).



- α denotes the amount of shrinkage.
- $\alpha = 0$ implies all features are considered and it is equivalent to the linear regression where only the residual sum of squares is considered to build a predictive model
- $\alpha = \infty$ implies no feature is considered i.e, as λ closes to infinity it eliminates more and more features

Lasso Regression

Key Idea: Lasso Regression minimizes residuals and adds a penalty on the sum of absolute coefficients ($|\theta_j|$).

Cost Function:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}_i - y_i)^2 + \lambda \sum_{j=1}^n |\theta_j|$$

Where:

- λ controls the penalty strength.
- Lasso can set some coefficients to zero, performing feature selection.

Lasso Regression - Effect of λ

- $\lambda = 0$: Equivalent to simple linear regression.
- $\lambda \rightarrow \infty$: All coefficients become zero, leading to underfitting.

Example Data:

x	y
1	2
2	2.5
3	3.5

Lasso helps with feature selection by shrinking some coefficients to zero.

Lasso Regression Example

Cost Function:

$$J(w_1, b) = \frac{1}{m} \sum_{i=1}^m \left(h(x^{(i)}) - y^{(i)} \right)^2 + \lambda |w_1|$$

Resulting Parameters:

$$w_1 = 0.6, \quad b = 1.47$$

Lasso Equation:

$$y = 0.6x + 1.47$$

Elastic Net Regression

Elastic net is a popular type of regularized linear regression that combines two popular penalties, specifically the L1 and L2 penalty functions.

Similar to Ridge and LASSO Elastic Net is an extension of linear regression, this adds regularization penalties to the loss function during training.

The penalty is a mixture of L1 and L2 penalties. Alpha and l1_ratio arguments decides the same. Default value of alpha is 1 and can be passed according to convenience of the data.

l1_ratio can have value from 0 to 1. Default value = 0.5. l1_ratio = 0 means the penalty is an L2 penalty. l1_ratio = 1 means it is an L1 penalty. l1_ratio is between 0 and 1, the penalty is a combination of L1 and L2.

Elastic Net Regression

Elastic Net Regression is a regularized regression technique that combines the strengths of both Ridge Regression (L2 penalty) and Lasso Regression (L1 penalty).

It is particularly useful when dealing with datasets that have:

- High dimensionality (many features, sometimes more than observations).
- Correlated predictors (where Ridge performs well but Lasso struggles).
- A need for both feature selection (Lasso) and coefficient shrinkage (Ridge).

The Elastic Net regression minimizes the following cost function:

$$\text{Loss} = \text{RSS} + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

Where:

- **RSS** = Residual Sum of Squares (error term).
- λ_1 = L1 regularization parameter (Lasso penalty).
- λ_2 = L2 regularization parameter (Ridge penalty).
- β_j = regression coefficients.

By tuning the parameters λ_1 and λ_2 , Elastic Net can balance between Lasso (sparse model with some coefficients set to zero) and Ridge (all coefficients shrunk but non-zero).

Key Advantages

- a. Handles multicollinearity better than Lasso.
- b. Performs variable selection like Lasso, but more stable.
- c. Works well with high-dimensional data (more features than observations).
- d. Combines **shrinkage + selection**, improving predictive performance.

THANK YOU !!